Homework 04 – Solution

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1 Setup

```
library(tidyverse)
library(readxl)

parlgov_election <- read_excel("parlgov.xlsx", sheet = "election")
parlgov_party <- read_excel("parlgov.xlsx", sheet = "party")
```

2 Exercise 1 – Import and Explore

2.1 Inspect the Data

```
glimpse(parlgov_election)
```

```
Rows: 9,016
Columns: 16
$ country_name_short               <chr> "AUS", "AUS", "AUS", "AUS", "AUS", "AU~
$ country_name                     <chr> "Australia", "Australia", "Australia",~
$ election_type                    <chr> "parliament", "parliament", "parliamen~
$ election_date                    <chr> "1901-03-30", "1901-03-
30", "1901-03-3~
$ vote_share                       <dbl> 44.4, 34.2, 19.4, 1.4, 0.6, 29.7, 34.4~
$ seats                            <dbl> 32, 26, 15, 1, 1, 26, 25, 23, 1, 0, 27~
$ seats_total                      <dbl> 75, 75, 75, 75, 75, 75, 75, 75, 75, 75~
$ party_name_short                 <chr> "PP", "FTP", "ALP", "none", "one-
seat"~
$ party_name                       <chr> "Protectionist Party", "Free Trade Par~
$ party_name_english               <chr> "Protectionist Party", "Free Trade Par~
$ left_right                       <dbl> 7.4000, 6.0000, 3.8833, NA, NA, 7.4000~
$ country_id                       <dbl> 33, 33, 33, 33, 33, 33, 33, 33, 33, 33~
$ election_id                      <dbl> 731, 731, 731, 731, 731, 730, 730, 730~
$ previous_parliament_election_id <dbl> NA, NA, NA, NA, NA, 731, 731, 731, 731~
$ previous_cabinet_id              <dbl> NA, NA, NA, NA, NA, 997, 997, 997, 997~
$ party_id                         <dbl> 1898, 1938, 1253, 1396, 2299, 1898, 19~
```

```
head(parlgov_election)
```

```
# A tibble: 6 x 16
  country_name_short country_name election_type election_date vote_share seats
  <chr>              <chr>        <chr>         <chr>              <dbl> <dbl>
1 AUS                Australia    parliament    1901-03-30          44.4    32
2 AUS                Australia    parliament    1901-03-30          34.2    26
3 AUS                Australia    parliament    1901-03-30          19.4    15
4 AUS                Australia    parliament    1901-03-30           1.4     1
5 AUS                Australia    parliament    1901-03-30           0.6     1
6 AUS                Australia    parliament    1903-12-16          29.7    26
# i 10 more variables: seats_total <dbl>, party_name_short <chr>,
#   party_name <chr>, party_name_english <chr>, left_right <dbl>,
#   country_id <dbl>, election_id <dbl>, previous_parliament_election_id <dbl>,
#   previous_cabinet_id <dbl>, party_id <dbl>
```

```r
summary(parlgov_election)
```

```
 country_name_short country_name       election_type      election_date     
 Length:9016        Length:9016        Length:9016        Length:9016       
 Class :character   Class :character   Class :character   Class :character  
 Mode  :character   Mode  :character   Mode  :character   Mode  :character  
                                                                            
                                                                            
                                                                            
                                                                            
   vote_share        seats          seats_total    party_name_short  
 Min.   : 0.00   Min.   :  0.00   Min.   :  5.0   Length:9016       
 1st Qu.: 2.10   1st Qu.:  1.00   1st Qu.: 81.0   Class :character  
 Median : 6.50   Median :  6.00   Median :152.0   Mode  :character  
 Mean   :11.79   Mean   : 23.21   Mean   :213.5                     
 3rd Qu.:17.50   3rd Qu.: 24.00   3rd Qu.:265.0                     
 Max.   :71.20   Max.   :470.00   Max.   :736.0                     
 NA's   :560     NA's   :64                                         
  party_name        party_name_english   left_right      country_id  
 Length:9016        Length:9016        Min.   :0.000   Min.   : 1.0  
 Class :character   Class :character   1st Qu.:3.300   1st Qu.:21.0  
 Mode  :character   Mode  :character   Median :5.714   Median :37.0  
                                       Mean   :5.146   Mean   :37.2  
                                       3rd Qu.:7.240   3rd Qu.:55.0  
                                       Max.   :9.825   Max.   :75.0  
                                       NA's   :947                   
  election_id     previous_parliament_election_id previous_cabinet_id
 Min.   :   1.0   Min.   :   1.0                  Min.   :   2.0     
```

```
 1st Qu.: 306.0   1st Qu.: 293.0                                1st Qu.: 361.0
 Median : 604.0   Median : 581.0                                Median : 752.0
 Mean   : 590.1   Mean   : 562.8                                Mean   : 786.3
 3rd Qu.: 866.0   3rd Qu.: 804.0                                3rd Qu.:1156.0
 Max.   :1124.0   Max.   :1122.0                                Max.   :1696.0
                  NA's   :297                                   NA's   :283
    party_id
 Min.   :   2
 1st Qu.: 535
 Median :1030
 Mean   :1142
 3rd Qu.:1577
 Max.   :2909
```

The election sheet has 9,016 rows and 16 columns. Each row represents one party's result in one election – so a single election with five parties competing produces five rows. Most columns are numeric or character; `election_date` is stored as a character string (format `"YYYY-MM-DD"`), and several columns such as `left_right` contain missing values (`NA`).

```r
# Distinct countries
parlgov_election |> distinct(country_name)
```

```
# A tibble: 37 x 1
   country_name
   <chr>
 1 Australia
 2 Austria
 3 Belgium
 4 Bulgaria
 5 Canada
 6 Switzerland
 7 Cyprus
 8 Czech Republic
 9 Germany
10 Denmark
# i 27 more rows
```

```r
# Observations by election type
parlgov_election |> count(election_type)
```

```
# A tibble: 2 x 2
  election_type     n
  <chr>         <int>
1 ep             1702
2 parliament     7314
```

There are 37 distinct countries in the dataset. Of the 9,016 rows, 7,314 (about 81%) come from national parliament elections and 1,702 from European Parliament elections. National elections are more numerous because they occur more frequently and go further back in time.

2.2 Subset to Parliament Elections

```r
parl <- parlgov_election |>
  filter(election_type == "parliament")

nrow(parl)
```

```
[1] 7314
```

After filtering to parliamentary elections only, 7,314 rows remain. This matches the count from the `count()` output above.

3 Exercise 2 – Filter and Select

3.1 German Elections

```r
germany <- parl |>
  filter(country_name == "Germany")

germany |> distinct(election_date)
```

```
# A tibble: 29 x 1
   election_date
   <chr>
 1 1919-01-19
```

```
 2 1920-06-06
 3 1924-05-04
 4 1924-12-07
 5 1928-05-20
 6 1930-09-14
 7 1932-07-31
 8 1932-11-06
 9 1933-03-05
10 1949-08-14
# i 19 more rows
```

The ParlGov data contains results from 29 German national parliamentary elections, ranging from January 1919 to the most recent election in the dataset.

3.2 Recent Western Europe

```r
recent_we <- parl |>
  mutate(year = as.numeric(str_sub(election_date, 1, 4))) |>
  filter(
    country_name %in% c("France", "Germany", "Netherlands", "Sweden"),
    year >= 2000
  ) |>
  select(
    country_name,
    election_date,
    party_name_english,
    vote_share,
    seats,
    seats_total,
    left_right
  )

recent_we
```

```
# A tibble: 269 x 7
   country_name election_date party_name_english    vote_share seats seats_total
   <chr>        <chr>         <chr>                      <dbl> <dbl>       <dbl>
 1 Germany      2002-09-22    Social Democratic Pa~       38.5   251         603
 2 Germany      2002-09-22    Christian Democratic~       29.5   190         603
 3 Germany      2002-09-22    Christian Social Uni~        9      58         603
```

```
 4 Germany      2002-09-22    Alliance 90 / Greens            8.6    55          603
 5 Germany      2002-09-22    Free Democratic Party           7.4    47          603
 6 Germany      2002-09-22    PDS | The Left                  4       2          603
 7 Germany      2002-09-22    The Republicans                 0.6     0          603
 8 Germany      2005-09-18    Social Democratic Pa~          34.2   222          614
 9 Germany      2005-09-18    Christian Democratic~          27.8   180          614
10 Germany      2005-09-18    Free Democratic Party           9.8    61          614
# i 259 more rows
# i 1 more variable: left_right <dbl>
```

The result contains 269 rows, covering national parliamentary elections in France, Germany, the Netherlands, and Sweden from 2000 onwards. Each row is one party result in one election, which looks correct: the four countries have had multiple elections over this period, each with several parties.

4 Exercise 3 – Mutate and Compute New Variables

4.1 Seat Share

```
parl <- parl |>
  mutate(seat_share = seats / seats_total * 100)
```

```
parl |>
  filter(!is.na(seat_share)) |>
  arrange(desc(seat_share)) |>
  select(country_name, election_date, party_name_english, seat_share) |>
  slice_head(n = 5)
```

```
# A tibble: 5 x 4
  country_name   election_date party_name_english                       seat_share
  <chr>          <chr>         <chr>                                         <dbl>
1 Canada         1958-03-31    Progressive Conservative Party of Can~         78.5
2 United Kingdom 1931-10-27    Conservatives                                  76.4
3 Canada         1984-09-04    Progressive Conservative Party of Can~         74.8
4 Canada         1940-03-26    Liberal Party of Canada                        73.9
5 Greece         1974-11-17    New Democracy                                  73.3
```

The five highest single-election seat shares all come from countries with majoritarian or first-past-the-post electoral systems – Canada, the United Kingdom, and Greece. In these systems, a party can win a large majority of seats with a vote share in the low-to-mid fifties. This illustrates the disproportionality that majoritarian systems produce, which we examine directly in the next task.

4.2 Vote–Seat Disproportionality

```
parl |>
  mutate(disproportionality = seat_share - vote_share) |>
  filter(!is.na(disproportionality)) |>
  group_by(country_name) |>
  filter(n() >= 50) |>
  summarise(
    mean_disprop = round(mean(disproportionality), 2),
    n = n(),
    .groups = "drop"
  ) |>
  arrange(desc(mean_disprop))
```

```
# A tibble: 37 x 3
   country_name mean_disprop     n
   <chr>               <dbl> <int>
 1 Bulgaria             0.48   132
 2 New Zealand          0.44   189
 3 Portugal             0.39   100
 4 Greece               0.35   138
 5 Latvia               0.29   107
 6 Turkey               0.29    76
 7 Estonia              0.28    79
 8 Spain                0.28   197
 9 Belgium              0.26   322
10 Slovakia             0.21   107
# i 27 more rows
```

The countries with the highest average disproportionality – Bulgaria, New Zealand (historically), Portugal, Greece – are or were predominantly majoritarian or mixed systems. In these cases, winning parties tend to receive seat bonuses: their seat share systematically exceeds their vote share. New Zealand's position near the top reflects its pre-1996 first-past-the-post system; after switching to mixed-member proportional representation the disproportionality dropped substantially, though the historical data still pulls the average up. Countries with low or negative mean disproportionality tend to use proportional representation systems, where seat shares track vote shares more closely.

5 Exercise 4 – Summarise and Group

5.1 Mean Left-Right Score by Country

```
parl |>
  filter(!is.na(left_right)) |>
  group_by(country_name) |>
  summarise(
    mean_lr = round(mean(left_right, na.rm = TRUE), 2),
    sd_lr = round(sd(left_right, na.rm = TRUE), 2),
    n = n(),
    .groups = "drop"
  ) |>
  filter(n >= 30) |>
  arrange(mean_lr)
```

```
# A tibble: 37 x 4
   country_name mean_lr sd_lr     n
   <chr>          <dbl> <dbl> <int>
 1 Portugal        3.88  2.41   142
 2 Spain           4.58  2.26   176
 3 Greece          4.63  2.64   138
 4 Iceland         4.68  2.08   180
 5 Slovenia        4.72  1.98   107
 6 Ireland         4.74  2.07   174
 7 Norway          4.78  2.39   238
 8 Switzerland     4.85  2.62   344
 9 Luxembourg      4.91  2.49   145
10 Japan           4.93  2.59   186
# i 27 more rows
```

The variation in mean left-right scores across countries partly reflects which parties have historically competed successfully in each system. Countries like Portugal and Spain – which transitioned to democracy in the mid-1970s after authoritarian regimes

– show lower mean left-right scores, consistent with the relative strength of left and centre-left parties in their post-transition party systems. The standard deviation column is arguably more substantively interesting: countries with high `sd_lr` have wider ideological spread across their competing parties, a basic indicator of party system polarisation.

5.2 Electoral Volatility Over Time

```
parl |>
  mutate(
    year = as.numeric(str_sub(election_date, 1, 4)),
    decade = (year %/% 10) * 10
  ) |>
  group_by(country_name, election_date, decade) |>
  summarise(max_vote = max(vote_share, na.rm = TRUE), .groups = "drop") |>
  group_by(decade) |>
  summarise(
    mean_max_vote = round(mean(max_vote, na.rm = TRUE), 1),
    n_elections = n(),
    .groups = "drop"
  ) |>
  filter(n_elections >= 5) |>
  arrange(decade)
```

```
# A tibble: 13 x 3
   decade mean_max_vote n_elections
    <dbl>         <dbl>       <int>
 1   1900          45.4          27
 2   1910        -Inf            40
 3   1920          36.9          57
 4   1930          39.4          48
 5   1940          40.5          49
 6   1950          40.7          66
 7   1960          41.1          59
 8   1970          38.6          73
 9   1980        -Inf            76
10   1990        -Inf           105
11   2000          34.7          98
12   2010          32.6         110
13   2020          30.6          37
```

The mean vote share of the largest party per election has declined steadily from around 45% in the 1900s to about 31% in the 2020s. This is consistent with the comparative politics literature on party system fragmentation: traditional mass parties have lost their dominance, new parties (particularly on the right and among Greens) have entered and split the vote, and voters have become less attached to established parties. The decline is not monotone – there is a slight uptick in the 1940s–1960s – but the long-run trend toward more fragmented electoral competition is clear.

6 Exercise 5 – A Full Pipeline

First, join party family information into the election data:

```
parl_with_family <- parl |>
  left_join(
    parlgov_party |> select(party_id, family_name),
    by = "party_id"
  )
```

Now the full pipeline:

```
parl_with_family |>
  mutate(year = as.numeric(str_sub(election_date, 1, 4))) |>
  filter(
    country_name %in%
      c(
        "France",
        "Germany",
        "Italy",
        "Spain",
        "Netherlands",
        "Belgium",
        "Austria",
        "Sweden",
        "Denmark",
        "Norway",
        "Finland",
        "Portugal"
      ),
    year >= 1990,
    !is.na(seats),
```

```r
    !is.na(family_name)
  ) |>
  group_by(family_name) |>
  summarise(
    total_seats = sum(seats, na.rm = TRUE),
    n_parties = n_distinct(party_name_english),
    .groups = "drop"
  ) |>
  arrange(desc(total_seats))
```

```
# A tibble: 10 x 3
   family_name         total_seats n_parties
   <chr>                     <dbl>     <int>
 1 Social democracy          10716        28
 2 Conservative               6579        34
 3 Christian democracy        4631        27
 4 Liberal                    4243        46
 5 Right-wing                 2414        37
 6 Communist/Socialist        2094        34
 7 Green/Ecologist            1438        22
 8 Agrarian                    930         4
 9 no family                   442         5
10 Special issue               223        39
```

Social democratic parties have accumulated by far the most seats in Western European parliaments since 1990, followed by Conservatives and Christian democrats. This broadly matches expectations: centre-left and centre-right parties have historically dominated Western European legislatures, even as their combined seat share has eroded with the rise of Green, populist right, and other challenger parties. The large number of distinct parties within the Conservative and Liberal families reflects the cross-national diversity of party labelling – parties that share an ideological family are grouped together regardless of their national names.

7 Exercise 6 – A Labelled Figure

This figure shows the mean vote share of Green and Ecologist parties in national parliamentary elections across Western European countries since 2000. Only countries where the Green family won at least one seat in at least one election are included.

```r
parl_with_family |>
  mutate(year = as.numeric(str_sub(election_date, 1, 4))) |>
  filter(
    family_name == "Green/Ecologist",
    year >= 2000,
    country_name %in%
      c(
        "France",
        "Germany",
        "Italy",
        "Spain",
        "Netherlands",
        "Belgium",
        "Austria",
        "Sweden",
        "Denmark",
        "Norway",
        "Finland",
        "Portugal",
        "Switzerland",
        "Ireland",
        "Luxembourg"
      )
  ) |>
  group_by(country_name) |>
  summarise(
    mean_vote = mean(vote_share, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  filter(n >= 2) |>
  mutate(country_name = fct_reorder(country_name, mean_vote)) |>
  ggplot(aes(x = mean_vote, y = country_name)) +
  geom_col(fill = "#4a7c59") +
  geom_text(
    aes(label = paste0(round(mean_vote, 1), "%")),
    hjust = -0.15,
    size = 3
  ) +
  scale_x_continuous(limits = c(0, 18), expand = c(0, 0)) +
  theme_bw() +
  labs(
```

```
    x = "Mean Vote Share (%)",
    y = NULL
  )
```

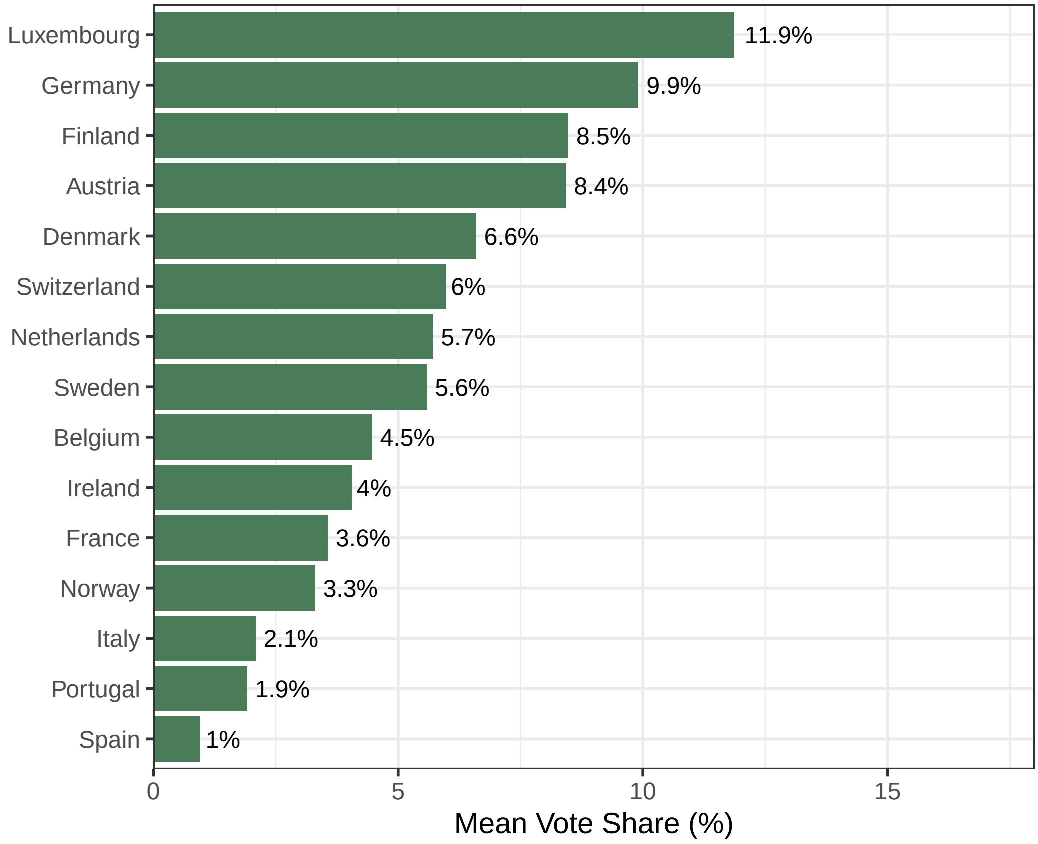

Figure 1: Mean vote share of Green/Ecologist parties in Western Europe, 2000–present

Figure 1 reveals substantial cross-national variation in Green party support across Western Europe since 2000. Luxembourg and Switzerland stand out as the strongest performers, with mean vote shares above 10%, reflecting both their proportional electoral systems and political cultures that have been receptive to post-materialist politics for several decades. Germany's Greens, among the oldest and most institutionalised Green parties in Europe, also score highly. At the lower end, France and Portugal show comparatively modest Green performance – in France's case, partly an artefact of the two-round majoritarian system that structurally disadvantages smaller parties. One limitation of this visualisation is that it averages across all elections since 2000, masking the strong upward trend in Green support that most countries experienced in the late 2010s following increased salience of climate change as a political issue.

8 Exercise 7 – Reflection